Dylan Dsouza

San Diego, CA
dydsouza@ucsd.edu | dylandsouza.com | linkedin.com/in/dsouza-dylan | github.com/dsouza-dylan

Education

University of California San Diego — San Diego, CA
Bachelor of Science in Data Science – GPA: 3.92 (Provost Honors) — *Expected March 2027*

Experience

Data Engineer Intern — Jul 2026 – Aug 2026
Capgemini — *Mumbai, India*
- Built a serverless AWS RAG pipeline for PDF Q&A using S3 storage, Titan embeddings via Bedrock, FAISS indexing, Lambda-parallelized retrieval, and Claude Sonnet responses via API Gateway — deployed to EC2 using CodePipeline.
- Completed a 15-course generative AI certification program covering prompt engineering, MCP protocol, agentic AI workflow design, RAG and vector database architectures, and AI governance.

AI Engineer — Oct 2025 – Jun 2026
Computer Science & Engineering Society — *San Diego, CA*
- Implemented a multi-stage Neo4j retrieval pipeline fusing MiniLM vector search and BM25 via reciprocal rank fusion, achieving 89% recall and 84% precision for an agentic AI platform modeling professional decision-making styles.
- Constructed a knowledge graph pipeline extracting LLM-generated concepts from research documents, mapping entity relationships via Girvan-Newman community detection across 8+ entity categories, and built the React/Vite frontend.

Machine Learning Engineer — Oct 2025 – May 2026
Engineers for Exploration — *San Diego, CA*
- Engineered a Google Earth Engine pipeline staging Sentinel-2 optical bands fused with 64-band AlphaEarth embeddings across training tiles in Madagascar and Mozambique, restricting downloads to mangrove regions via ESA WorldCover.
- Fine-tuned a continual learning segmentation model on 21,000+ GeoTIFF image-mask pairs, achieving 82–85% mIoU across 4 land-cover classes for global-scale pixel-level mangrove mapping.

Data Analyst — May 2025 – May 2026
Scripps Institution of Oceanography — *San Diego, CA*
- Automated 11 years of coral reef data processing (60% cut in manual processing time), building a Python ETL pipeline that standardized 40,000+ benthic annotations into clean datasets for downstream analysis across 15 reef sites.
- Applied EDA and time series analyses across 5 experimental groups and 16 survey time points, finding active restoration plots reached 30–40% coral cover versus 13% in controls — quantifying the long-term impact of transplantation.

Projects

see all projects →

Who Gets the Old Planes? | *Personal Project* — View Project | GitHub
- Cross-referenced 2.7M flights with FAA aircraft registration records to map narrowbody fleet age across U.S. airports, uncovering an 18-year disparity between the oldest and youngest fleets driven by market competition, not airline.

DegreeDelta: Quantifying Education Returns | *Course Project (Statistical Methods)* — View Project | GitHub
- Quantified the college income premium across all 50 U.S. states using 3.4M weighted ACS PUMS records, revealing that cost-of-living adjustment dramatically reshuffles which states actually deliver the highest return on a degree.

ENSOcast: Climate Analytics Dashboard | *Conference Project (SDUTC 2025)* — View Project | GitHub
- Engineered a Streamlit climate dashboard on 40+ years of NOAA data; benchmarked Random Forest, XGBoost, 1D CNN, LSTM, and Ensemble models achieving 82% accuracy in 3-class ENSO phase prediction.

Predicting U.S. Power Outage Severity | *Personal Project* — View Project | GitHub
- Classified U.S. power outage duration as long or short across 16 years of continental data (2000–2016), engineering geographic, climatic, and infrastructure features to achieve 80% accuracy with a tuned Random Forest.

Skills

Languages: SQL (PostgreSQL), Python, R, JavaScript, HTML, CSS, Java
Libraries: pandas, NumPy, statsmodels, SciPy, Matplotlib, Seaborn, scikit-learn, XGBoost, NLTK, HF Transformers, FAISS
Frameworks: dbt, Airflow, Spark, Streamlit, pytest, MLflow, PyTorch, LangChain, FastAPI
Tools: Git, GitHub Actions, AWS, BigQuery, Tableau, Power BI, Docker, Excel, Kafka, Iceberg, Neo4j, Bash, Claude Code
Core Competencies: EDA, Statistical Modeling, Hypothesis Testing, Data Visualization, Machine Learning, Feature Engineering, Time Series, NLP, Deep Learning, GenAI, RAG, Vector Databases, Prompt Engineering, CI/CD, Data Journalism